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Upgrading financial education by adding Python-based personalized financial projection: A randomized control trial

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Abstract

Research has shown that even though standardized financial education has gained prevalence to promote financial literacy over the past decade, it has had little effect on personal financial planning. The present study used a randomized control trial to examine the effectiveness of a Python-based personalized financial projection on young working adults in Hong Kong, to examine if and how this approach improves their financial planning. Participants assigned to the experiment group received standardized financial education and Python-based financial projections, while those in the control group only received standardized financial education. The assessment based on the two-wave data showed that Python-based financial projection promoted future time perspectives, reduced temporal discounting, and improved financial planning via the full mediation of promoting financial attitudes. Although numerous applications for personal financial planning exist (such as Wallet, Walnut, Monefy, and Money View), our Python-based financial projection stands out as the pioneering solution tailored for the hands-on manipulation of programming code to effectively manage personal finances. Our findings suggest a new track to upgrade personalized financial projection and standardized financial education and contribute generously to the development of personal finance education.

KEYWORDS
financial planning, personal finance education, personalized financial projection, Python, standardized financial education, temporal discounting

Practitioner notes

What is already known about this topic

- Standardized financial education promotes objective financial knowledge.
- Standardized financial education has a limited effect on personal financial planning.
- Classical personalized financial projection promotes personal financial planning, but the effect is small.

What this paper adds

- Introduction of a novel Python-based personalized financial projection by manipulating projection code.
- The evidence that Python-based personalized financial projection more strongly improves personal financial planning, compared to the classical personalized financial projection.
- The evidence why Python-based personalized financial projection can improve personal financial planning.

Implications for practice and/or policy

- Facilitating engagement of young working adults with personalized finance planning through the use of a Python-based intervention.
- Integrating Python-based personalized financial projection into standardized financial education in the school setting.
- Using Python as the platform to design more topic-specific financial education module.

INTRODUCTION

There has been a surge in access to financial technology and diversified financial products and services in modern societies, such as Hong Kong. The rise in financial responsibility among individual consumers necessitates the equipment of basic financial literacy. Recently, the personal finance literature described financial literacy as a multidimensional construct rather than objective financial knowledge alone (Hizgilov & Silber, 2020; Lyons & Kass-Hanna, 2021). Consumers with high financial literacy refer to those who develop a sound understanding of personal finance terms, hold positive attitudes toward healthy financial habits, have self-esteem in financial practice and actively engage in financial planning (Rai et al., 2019; Xiao & O'Neill, 2016). Understanding financial literacy as a multidimensional construct carries significant importance because personal financial outcomes are influenced by a combination of factors, including financial knowledge, attitudes, confidence, planning, and decision-making (Douissa, 2020; Hizgilov & Silber, 2020). Isolating any one of these factors overlooks the intricate interplay between them and their collective impact on financial well-being. This multidimensional view has a profound impact on how financial literacy should be assessed, and places heightened emphasis on the necessity for financial education programs to be comprehensive.

Standardized financial education has gained prevalence among practitioners in the last decade to promote financial literacy (Amagir et al., 2018). However, it has little effect on personal financial planning, although it improves objective financial knowledge (Kaiser & Menkhoff, 2020). The main limitation of a generic intervention is that it cannot be flexibly adjusted to fit an individualized financial status and needs. On the other hand, *personalized* financial intervention is more potent in shaping financial planning because it can affect participants' underlying psychology (Bartels & Urminsky, 2015; Wiener & Doescher, 2008). In particular, personalization harnesses the power of intrinsic motivation by aligning with an individual's personal values and aspirations (Li et al., 2020, 2023). When financial advice deeply resonates with a person's goals, it sparks a heightened motivation to actively engage in personal financial management. Furthermore, the act of tailoring advice to suit each individual's distinct circumstances fosters a robust sense of self-efficacy (Li et al., 2020). This, in turn, empowers them to embark on proactive measures within their financial planning journey.

Behavioural scientists have developed psychological models that explain why people actively join in personal financial planning (Serido et al., 2013; Tomar et al., 2021). Two models were tested and validated by our team in Hong Kong using data collected from adolescents and working adults (Zhu, 2018; Chou et al., 2015). Our results suggest that future perspective, temporal discounting, and financial attitudes play significant mediational roles in the pathway toward skilful financial planning. There are hierarchical relationships among future time perspectives, temporal discounting, and financial attitudes. Future time perspectives only capture attitudes toward the future, while temporal discounting reflects intertemporal choices (ie, how a consumer will choose between the present and the future) (Chapman, 1996; Seaman et al., 2022). When consumers perceive that the future is more valuable than the present (ie, low temporal discounting), there will be a willingness to save and invest for the future and to engage in budget and consumption control in the present (ie, positive financial attitudes). Therefore, a personalized financial intervention manipulating future time perspectives, temporal discounting, and financial attitudes may promote the skills and intentions of financial planning.

Temporal discounting theory suggests that people normally devalue the future because of their psychological disconnectedness from the future (Bartels & Urminsky, 2015). Without cognitive intervention, people can hardly establish the vividness of and a connection to the future self, not to mention sacrificing immediate temptation for the benefit of the future self (eg, controlling immediate spending and saving for the future). That is why behavioural scientists resort to computer-based interventions to promote the vividness of the future, such as age-progressed renderings with virtual reality technologies, discussion with future self-produced AI technology, and remarkably personalized financial projection with simulation techniques (Goda et al., 2014; Hershfield et al., 2011; Marques et al., 2018).

Recently, our team conducted a three-arm experiment to examine the effectiveness of a computer-based personalized financial projection among working adults in Hong Kong. The projection simulates the financial resources accumulated in the distant future based on a working adult's current earnings, savings and investment behaviours (Zhu et al., 2023). We masked the working mechanism of the projection and created a mobile application that showed the projected financial outcomes to the participants after they input their current financial choices. The experimental results showed that the personalized financial projection significantly reduced temporal discounting, but the effect was small ($b^* = -0.11$). The findings also revealed that the projection did not significantly shape future time perspectives. Both limitations are likely due to a direct connection between the present and the future, without vivid guidance on how the present and future selves are connected (Oyserman et al., 2004). One potential solution could involve introducing participants to the coding of a computer-based personalized financial projection. This approach would grant them the opportunity to manipulate projection code, thus enabling the forecasting of individualized financial trajectories. This process would serve to elucidate the intricate interplay and connection between the present and the future.

Python, created by Guido van Rossum and released in 1991, was recognized as the most popular coding tool in the world by the Popularity of Programming Language Index (PYPI, 2020). Python has been observed to have a low threshold for learners and powerful computing, design and simulation functions. Humber (2018) adopted Python as a coding tool to simulate personal financial planning. Considering that Python is user-friendly, in this study, we used Python as the coding language to develop a personalized financial projection program. Specifically, we created an experimental condition in which we taught young working adults basic Python grammatical rules and how to manipulate projection code developed by us to create "their" financial projections. The primary objective of this study was to test whether Python-based personalized financial projection could achieve the following:

1. More strongly reduce temporal discounting compared to our previous financial projection, which masks the coding.
2. Promote future time perspectives and financial attitudes.
3. Improve financial planning via the mediation of temporal discounting, future time perspectives and financial attitudes.

LITERATURE REVIEW

The motivation for upgrading standardized financial education

There is a shared sense of lifelong financial education that educators should start introducing basic financial knowledge in primary schools, delivering standardized and comprehensive financial education in secondary schools, and offering topic-specific financial training among college students and working adults (Amagir et al., 2018; Garcia, 2013; Walstad et al., 2017). Financial education has even been extended to older adults to assist them in managing their healthcare choices (MacLeod et al., 2017).

By the end of 2017, more than 70 countries had developed national strategies to comprehensively promote the financial literacy of the next generation of young adults (OECD, 2017). Educational authorities have added standardized financial training to the secondary-level curriculum structure in 15 of the 17 Asia-Pacific Economic Cooperation (APEC) economies (OECD, 2019). Considering that there has been no standardized financial education in secondary schools in Hong Kong, our team introduced an international standardized financial curriculum, the Financial Fitness for Life (FFFL), to Hong Kong adolescents and conducted the first randomized control trial in Asia to assess its impacts on multiple dimensions of financial literacy (Zhu, 2020; Zhu et al., 2021). The FFFL was developed by the Council for Economic Education in the U.S. and was appraised as the most comprehensive standardized financial education in the international community (Batty et al., 2015; Berry et al., 2018; Walstad et al., 2010). The FFFL teaches participants basic concepts to make sound decisions regarding earning income, spending, saving, borrowing, investing and managing money.

Our experiment showed that the FFFL was sufficiently powerful to enhance the objective financial knowledge of participants but inadequate to bring positive change to their financial planning (ie, saving, budgeting and spending control; Author, 2020; Author et al., 2021). Recent systematic reviews and meta-analyses consistently suggest that standardized financial education is more potent in shaping objective financial knowledge than financial planning behaviours (Amagir et al., 2018; Kaiser & Menkhoff, 2020). Hence, this paper proposes intelligent solutions to upgrade standardized financial education to promote participants' financial planning. In this study, we designed and conducted a Python-based personalized financial projection and examined its intervention mechanism (ie, how it promoted financial planning for participants by changing future time perspectives, temporal discounting and financial attitudes).

Python-based personalized financial projection, future time perspectives and temporal discounting

Individuals experience multiple events that break the life course into several independent sections (Rutt & Löckenhoff, 2016). Typical events include going to college as a freshman, getting married, studying overseas and relocating family. People may perceive these events as turning points that reduce the discontinuity of the whole life course (Hershfield & Bartels, 2018). When there is inadequate continuity between the present and the future, people do not believe that the future quality of life is related to the present sacrifice. They would rather believe that these turning points (ie, life events) determine future well-being in a random walk. The future can never be valuable when vague, uncertain and obscure (Hershfield et al., 2011; Pronin & Ross, 2006).

Unlike financial projection without presenting coding details (eg, Fajnzylber & Reyes, 2015; Fuentes et al., 2022; Goda et al., 2014), Python-based financial projection can reduce temporal discounting more strongly and improve future time perspectives. Python-based coding links the present to the future by convincing users that current financial choices determine future financial outcomes, thereby increasing the perceived continuity of life courses. The coding presents a detailed simulation model illustrating how financial status and choice develop into future financial outcomes. The perceived value of the future automatically increases when the future is no longer an ambiguous object but a predictable outcome following a clear development track. Therefore, we propose the following hypotheses:

H1. Python-based financial projection reduces temporal discounting more strongly.

H2. Python-based financial projection promotes future time perspectives.

H3a & H3b. Python-based financial projection improves personal financial planning by the mediation of reducing temporal discounting and promoting future time perspectives, respectively.

Python-based personalized financial projection and financial attitudes

Experiential learning theory (ELT) constructs a holistic model that fits the learning process and human development in different fields (Kolb et al., 2014), such as financial literacy (composed of knowledge, attitudes and behaviours). People must experience two stages to develop sound financial literacy: grasping experience and transformation experience (Noh, 2022). The grasping experience contains substantial financial experience (eg, by frequent financial practice) and abstract concepts (eg, by standardized financial education). To gain transformation experience, learners must develop reflective observation (eg, by observing the financial practices of others) or active experimentation (eg, by trial and error).

Our previously validated model among youth in Hong Kong reveals that when people engage in a type of financial practice with high frequency (ie, substantial financial experience in the grasping experience of ELT), they develop financial habits without promoting corresponding financial attitudes (Zhu, 2018). Additionally, our previous experiment showed that standardized financial education (ie, abstract concepts in the grasping experience of

ELT) helped participants acquire financial concepts but failed to shape their positive financial attitudes (Zhu, 2020). Hence, scholars should seek solutions to improve financial attitudes by promoting a transformation experience of ELT (reflective observation or active experimentation).

The Python-based personalized financial projection design allows participants to be observers and practitioners simultaneously. Participants may adjust the parameters (eg, interest rate, inflation rate, investment rate of return) and the inputs in the program (eg, salary, age, expenses) to review the financial details of others and themselves. They may adjust these variables multiple times to identify the best planning strategy to achieve their ideal financial outcomes (eg, experimentation). In comparison, classical personalized financial projections do not allow for the adjustment of parameters and present a simple and friendly interactive window (eg, a mobile application) for participants to adjust inputs only (Fajnzylber & Reyes, 2015; Fuentes et al., 2022; Goda et al., 2014). Furthermore, the utilization of Python-based financial projection empowers participants with the capability to refine the projection process. This is achieved through the revision of pre-existing code or the incorporation of entirely new code elements, thereby fostering robust experimentations and explorations. In summary, Python-based financial projection motivates participants to "play" the model and immerse themselves in concrete realities to learn and develop financial literacy, which utilizes the core advantage of ELT. Therefore, we offer the following hypotheses:

H4. Python-based financial projection improves financial attitudes.

H5. Python-based financial projection promotes personal financial planning by the mediation of improved financial attitudes.

METHOD

General design

This study aimed to upgrade standardized financial education by adding a Python-based personalized financial projection. To justify this, we designed a randomized control trial to test how and to what extent the Python-based financial projection improved financial literacy. Participants in the experimental group received standardized financial education as well as Python-based financial projections, whereas participants in the control group solely received standardized financial education. This design enabled us to examine the unique effect of Python-based financial personalized projection on three mediators (ie, future time perspectives, temporal discounting and financial attitudes) and one outcome (ie, financial planning of participants).

Standardized financial education

We implemented FFFL modules to conduct standardized financial education. The FFFL has been validated among young people in Hong Kong (Zhu et al., 2021). Financial topics in the FFFL (ie, earning income, spending, saving, borrowing, investing and managing money) were seamlessly connected using the life course financial planning model (Salignac et al., 2020). For example, savings and investment are necessary to establish financial well-being throughout life, while budgeting and spending control are prerequisites to saving and engaging in investment. We integrated multiple teaching strategies into the two-hour training, including lectures, case studies, group discussions, videos, games and role-playing.

Python-based personalized financial projection

The Python-based personalized financial projection is the core component of our intervention. It is composed of five hours of training in basic Python grammatical rules, two hours of coding skills for beginners and applying two financial projection models to facilitate financial planning. This intervention entails the manipulation of projection code developed by our team to cater to individualized circumstances. Mosh (2022) developed a short course to familiarize beginners with Python skills in around 10 hours. We used it to train the participants in Python grammar, but shortened it to five hours because we needed to cover coding skills related to two financial projection models only. The details are reported in Table 1.

Bearing in mind that we aimed to improve the financial planning of participants by promoting their future time perspectives and weakening temporal discounting, we designed two financial projection models to more vividly demonstrate the time value of money. The first model (ie, the money management model) simulated an individual's earning, spending, saving and investment behaviours in the upcoming years to show that compound interest could appreciate wealth and build financial well-being in the future. With a foundational understanding of Python-based coding at their disposal, participants were encouraged to actively calibrate economic parameters (such as interest rates, inflation rates and investment returns) and individual inputs (including monthly expenditures and saving rates) in the projection code we developed to simulate the progressive accumulation of wealth. Furthermore, participants were allowed to introduce new code elements if they deemed it beneficial in mirroring their personalized financial trajectory. Paramount to this experience was the iterative nature of the model, which vividly underscored the far-reaching consequences of contemporary decisions on future quality of life. Beyond this, participants were notably encouraged to factor in the appreciation of money over time when arriving at financial decisions. The structural flow of the first model is visually expounded upon in Figure 1.

Next, we developed the second model (eg, debt management with a credit card) to help participants learn about the substantial future cost after falling into the “revolving credit” trap when using a credit card for consumption. The model aimed to effectively show that the total cost of consumption with revolving credit would be much higher than the original price of the commodity, and the considerable pressure of repayment may seriously lower the quality of life in the future. Analogous to the preceding model, participants were invited to try multiple scenarios by adjusting the economic parameters (eg, interest rates for borrowing, the most prolonged borrowing period) and personal inputs (eg, monthly payments, loan periods without a contract). Concurrently, participants were encouraged to refine existing code elements or even forge new ones if they believed such modifications would enhance the personalization of the projection. Significantly, these explorations underscored the pivotal link between present-day repayment actions and future financial well-being. The procedural trajectory of the second model is visually illustrated in Figure 2.

Participants

Python-based financial projection belongs to topic-specific financial education. According to the life course financial education landscape, it best fits the working adult group (Walstad et al., 2017). Young working adults are digital natives surrounded by computers and digital devices and maintain a shorter psychological distance with a programming tool such as Python. Therefore, we believe it is beneficial to limit the participants to young working adults to maximize the potential effect of our intervention. Considering that personal financial status is the fundamental condition for engaging in financial activities, there is a motivation to

TABLE 1 Basic Python coding.

Topic	Description	Time
Environment of program development	Installing PyCharm and running Python-based coding	1 hour
Variable	A symbolic name that is a pointer to an object	
Type of data	Integer, floating point, string, and bool	
Input function	Reading a line from the input (usually from the user)	
Transforming data type	Performing conversion from one data type to another	
Format String	Formatting the specified value(s) and insert them inside the string's placeholder	
Mathematical calculation and operators precedence rule	Addition (+), subtraction (−), division (/), multiplication (*), remainder (%), square (**), and integer division (//)	0.5 hour
The “if” statement	Deciding whether certain statements need to be executed or not.	0.5 hour
Logical operators	Logical AND, Logical OR and Logical NOT operations	0.5 hour
The “while” loop	Running a block code until a certain condition is met	0.5 hour
The “for” loop	Iterating over a sequence (eg, a list)	0.4 hour
List	A collection which is ordered and changeable	0.4 hour
Tuple	Storing multiple items in a single variable	0.4 hour
Dictionary	A hashed structure of various pairs of keys and values	0.4 hour
Functions	A block of code which only runs when it is called	0.4 hour

Note: The description was extracted from VanderPlas (2016).

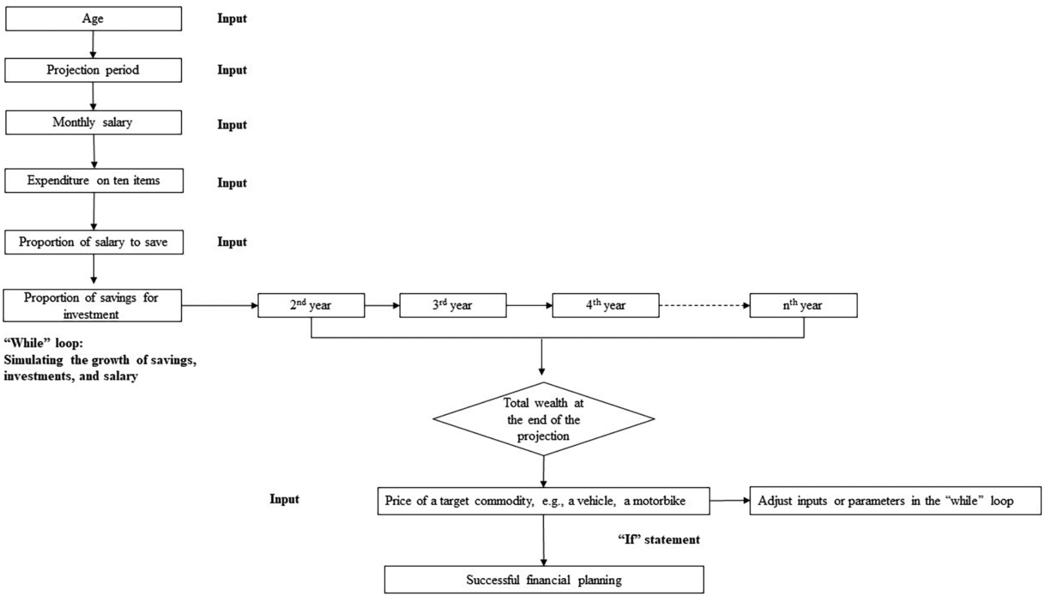

FIGURE 1 Model of the money management.

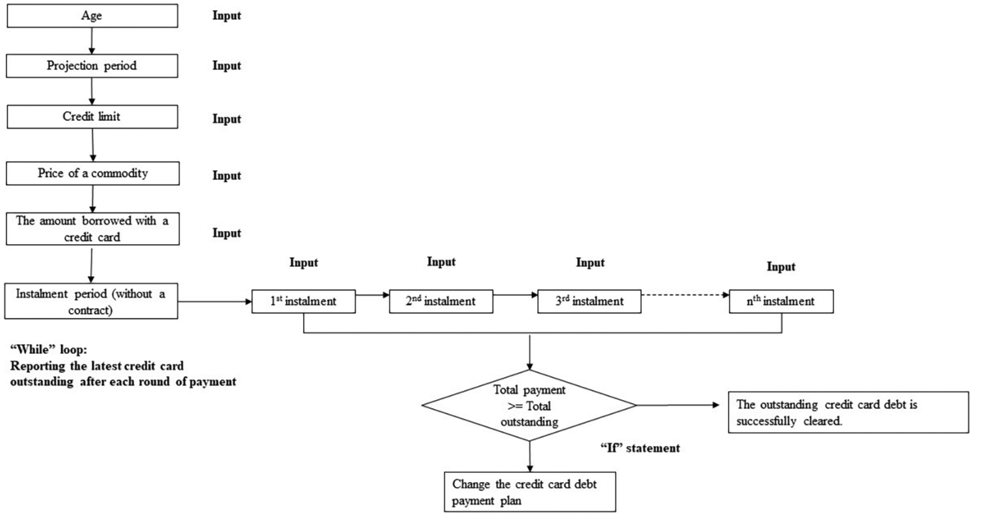

FIGURE 2 Model of the debt management with a credit card.

further limit the participant to those with solid earning power. Overall, the participant has to meet the following criteria:

1. Aged 18–35 years.
2. Full-time administrative staff at public universities in Hong Kong, with a stable monthly income above the median in Hong Kong.
3. Not holding a degree in finance or computer science.
4. Not having taken any standardized financial courses.
5. Not a Python coder.

Procedures

We emphasize the quantitative nature of our study, which hinges on the collection of first-hand data to rigorously evaluate a two-arm experimental model. The ensuing sections delineate the meticulous procedure we employed, elucidating our efforts to ensure robustness in our experimental design and data collection.

The research team appointed two part-time research officers. The first was responsible for recruiting participants, preparing assessment instruments and managing research data. The second staff member was assigned duties related to Python and FFFL (ie, preparing intervention materials and developing the Python-based financial projection program). We used "Wix.com (a free online website builder)" to create a public platform that published comprehensive information regarding our intervention, including indicative contents, our team members and the confirmed timetable. To fit the timetable of the university staff, we offered experimental conditions on Friday night and daytime on the following Saturday. The arrangement ensured that there was a low likelihood of attrition during the intervention. All teaching and learning activities were arranged online via Zoom meetings.

The email contacts of full-time administrative staff in eight publicly funded universities in Hong Kong are open. We created the registration link using Qualtrics XM and attached it to the mass email invitation. In the invitation, we detailed the research objective and the significance of this study in the personal finance landscape in Hong Kong and guaranteed that those who completed the training and two tests (ie, pretest and post-test) would get a supermarket cash coupon worth HK$100 (= US$12.90).

People who registered for the intervention received a pretest link. We randomly assigned participants who completed the pretest to either the experiment or the control group. For the experiment group, we offered the intervention in two rounds (ie, on the first and second weekends of January 2023). In each round, we accepted at most 25 participants to ensure adequate communication between the trainer and each trainee. We conducted the post-test after they completed both the FFFL and Python-based financial projections. The training in the experimental group lasted 9 hours in total.

In the control group, we only taught the FFFL module (ie, standardized financial education lasting for only 2 hours) between the pretest and post-test. The FFFL was delivered on the third weekend of January 2023. However, for ethical considerations, we offered the same Python-based financial projection to the control group participants after they completed the post-test. In other words, the Python-based financial projection in the control group was not part of our randomized control trial. It was only a gift to the participants in the control group.

Measurements

In the pretest and post-test, we adopted assessment instruments that have been validated among working adults in Hong Kong to measure potential mediators (ie, future time perspective, temporal discounting and financial attitudes) and one outcome variable (ie, financial planning) (Chou et al., 2015, Zhu et al., 2021, 2023). Future time perspectives were assessed using six items on a five-point scale (1 = highly disagree; 5 = highly agree) extracted from Tomar et al. (2021), with excellent internal consistency reliability in the pretest and post-test ($\alpha = 0.88$ and 0.90). The six items included the following:

- "I like to think about what the future will hold."
- "I enjoy thinking about how I will live years from now in the future."
- "I look forward to life in the distant future."
- "According to me, it is important to have a long-term perspective in life."

- "My close friend would describe me as future-oriented."
- "I look forward to a long journal toward the distant future."

We invited participants to play two games to assess temporal discounting (ie, making hypothetical choices between an immediate reward and a delayed fixed reward) (Basile & Toplak, 2015). In the first game, we asked participants to choose between HK$10,000 now and HK$10,000 one month later. We gradually reduced the amount that could be obtained now and created an additional 11 scenarios for participants to choose from two options. Once a turning point occurred (ie, participants switched to get the amount one month later), the present value of HK$10,000 one month later (ie, the reversed temporal discounting) was identified. In the second game, we repeatedly asked participants to make 12 hypothetical choices but adjusted the delayed period from one month to one year. The total 24 items in the two games are presented in Table 2. We identified the temporal discounting shortly and in the distant future by reviewing participants' choices.

Financial attitudes were assessed with three items developed and used in Shim et al.'s (2015) study. We asked participants to report their agreement with three financial behaviours on a five-point scale (1 = highly disagree; 5 = highly agree), including:

- "You should make sure there is a sufficient balance in your bank account."
- "You should clean the debts in your credit card every month."
- "You should save to prepare for emergencies."

The internal consistency across the three items was adequate (pretest: $\alpha = 0.87$; post-test: $\alpha = 0.85$). Financial planning was measured by a single item, "Please report your financial planning activities (Salignac et al., 2019)." The answers were calibrated on a four-point scale (1 = In the current financial status, it is impossible for me to engage in any financial planning activity; 4 = In the current financial status, it is effortless to plan for my finance).

In addition to the mediators and the outcome variable, we assessed the background variables, ie, educational achievement, monthly personal income, monthly family income, and total family assets. We asked participants to report their educational achievements by choosing from 1 (no formal schooling) to 6 (a postgraduate degree or above). Financial status is a critical prerequisite to financial planning, so it must be measured among participants and added as the control in main analysis. Considering that financial status was sensitive information to participants, we invited them to choose a range rather than report specific amounts when measuring monthly personal income, monthly family income, and total family assets. The results were as follows:

- Monthly personal income ranged from 1 (HK$18,700 − HK$19,999, UD$1 = HK$7.75) to 6 (HK$60,000 or above).
- Monthly family income ranged from 1 (HK$18,700 − HK$19,999) to 6 (HK$100,000 or above).
- Total family assets ranged from 1 (HK$0 − HK$ 499,999) to 11 (HK$5,000,000 or above).

Data analysis

Following the standardized procedures of experiment-based intervention research, we checked whether the group assignment (ie, assigning participants to either the experiment or control group) was adequately random by examining whether there was a significant difference in mediators, outcome variables and background variables (Becchetti et al., 2013; Kalwij et al., 2019). Specifically, we conducted an independent *t*-test to compare the means

TABLE 2 Items in the simulation games.

		Option 1	Option 2
1	Option 1: Get HK$ 10,000 now option 2: Get HK$ 10,000 one month later	1	2
2	Option 1: Get HK$ 9000 now Option 2: Get HK$ 10,000 one month later	1	2
3	Option 1: Get HK$ 8000 now Option 2: Get HK$ 10,000 one month later	1	2
4	Option 1: Get HK$ 7000 now Option 2: Get HK$ 10,000 one month later	1	2
5	Option 1: Get HK$ 6000 now Option 2: Get HK$ 10,000 one month later	1	2
6	Option 1: Get HK$ 5000 now Option 2: Get HK$ 10,000 one month later	1	2
7	Option 1: Get HK$ 4000 now Option 2: Get HK$ 10,000 one month later	1	2
8	Option 1: Get HK$ 3000 now Option 2: Get HK$ 10,000 one month later	1	2
9	Option 1: Get HK$ 2000 now Option 2: Get HK$ 10,000 one month later	1	2
10	Option 1: Get HK$ 1000 now Option 2: Get HK$ 10,000 one month later	1	2
11	Option 1: Get HK$ 500 now Option 2: Get HK$ 10,000 one month later	1	2
12	Option 1: Get HK$ 100 now Option 2: Get HK$ 10,000 one month later	1	2
13	Option 1: Get HK$ 10,000 now Option 2: Get HK$ 10,000 one year later	1	2
14	Option 1: Get HK$ 9000 now Option 2: Get HK$ 10,000 one year later	1	2
15	Option 1: Get HK$ 8000 now Option 2: Get HK$ 10,000 one year later	1	2
16	Option 1: Get HK$ 7000 now Option 2: Get HK$ 10,000 one year later	1	2
17	Option 1: Get HK$ 6000 now Option 2: Get HK$ 10,000 one year later	1	2
18	Option 1: Get HK$ 5000 now Option 2: Get HK$ 10,000 one year later	1	2
19	Option 1: Get HK$ 4000 now Option 2: Get HK$ 10,000 one year later	1	2
20	Option 1: Get HK$ 3000 now Option 2: Get HK$ 10,000 one year later	1	2
21	Option 1: Get HK$ 2000 now Option 2: Get HK$ 10,000 one year later	1	2
22	Option 1: Get HK$ 1000 now Option 2: Get HK$ 10,000 one year later	1	2
23	Option 1: Get HK$ 500 now Option 2: Get HK$ 10,000 one year later	1	2
24	Option 1: Get HK$ 100 now Option 2: Get HK$ 10,000 one year later	1	2

across two groups in age, educational achievement, monthly personal income, monthly family income, family assets, future time perspectives (mediator), temporal discounting shortly (mediator), temporal discounting in the distant future (mediator), financial attitudes (mediator) and financial planning (outcome variable). We used the Chi-square test to compare the male proportion differences between the experimental and control groups.

In the main analysis, we used multiple regression to investigate whether participants in the experimental group reported higher future time perspectives, lower temporal discounting, more positive financial attitudes and better financial planning than those in the control group when controlling for the baseline status of these focal constructs. In the next step, we added background variables to the regression model to validate the results.

Finally, we used the structural model to confirm whether the Python-based financial projection could promote the outcome variable (ie, financial planning) via four mediators: future time perspectives, temporal discounting shortly, temporal discounting in the distant future, and financial attitudes. In the structural model, we controlled for the status of the four mediators and the outcome in the pretest. Additionally, we linked the Python-based financial projection to the four mediators and the outcome measured in the post-test. Moreover, we added pathways from four mediators to the outcome variable in the post-test.

RESULTS

The Chi-square and t-tests showed no significant differences in all variables (mediators, outcome and background variables) between the experiment and the control group (see Table 3), suggesting that our group assignment was successful. The experimental results reported in Table 4 show that Python-based financial projection expectedly influenced all mediators and the outcome variable. Before controlling for background variables, the Python-based financial projection significantly promoted future time perspectives ($\beta = 0.18$, $p < 0.05$), financial attitudes ($\beta = 0.22$, $p < 0.05$), and financial planning ($\beta = 0.24$,

TABLE 3 Descriptive statistics in the experimental and control groups.

	Mean (standard deviation), %	
	Control group ($N = 17$)	**Experimental group ($N = 44$)**
Background variables		
Gender, being male (%)	46.7	32.6
Age	29.23 (4.69)	30.09 (7.87)
Educational achievement	5.47 (0.52)	5.39 (0.49)
Monthly personal income	2.57 (2.06)	2.73 (1.06)
Monthly family income	4.14 (2.21)	4.17 (1.96)
Family asset	2.62 (1.19)	2.35 (1.53)
Mediators		
Future time perspective	3.69 (0.61)	3.70 (0.71)
Temporal discounting shortly	667 (1371)	360 (762)
Temporal discounting in the distant future	1727 (1902)	1750 (1795)
Positive financial attitudes	4.26 (0.83)	4.20 (0.61)
Outcome variable		
Financial planning	2.86 (0.77)	3.24 (0.88)

Note: The Chi-square and independent sample *t*-test showed no significant difference in all variables.

TABLE 4 Results of multiple linear regressions.

	Future time perspective[a]		Temporal discounting shortly[a]		Temporal discounting in the distance future[a]	
Experimental group	0.18*	0.18*	−0.21	−0.31*	−0.27	−0.29*
Pretest	0.80***	0.81***	0.37*	0.37*	0.35*	0.10
Gender, being male (%)		−0.16*		−0.14		−0.16
Age		−0.01		0.25		0.34
Educational achievement		0.17*		0.06		−0.04
Monthly personal income		−0.12		−0.20		0.23
Monthly family income		−0.010		−0.39*		−0.66**
Family asset		0.10		0.36*		0.26
	Financial attitudes[a]		**Financial planning[a]**			
Experimental group	0.22*	0.21*	0.24*	0.19*		
Pretest	0.63***	0.66***	0.67***	0.65***		
Gender, being male (%)		−0.15		−0.09		
Age		0.28*		0.26*		
Educational achievement		−0.01		−0.05		
Monthly personal income		−0.28		−0.11		
Monthly family income		0.05		0.18		
Family asset		−0.07		−0.26*		

[a]Posttest scores. Standardized estimated coefficients are reported.
*$p<0.05$; **$p<0.01$; ***$p<0.001$.

$p<0.05$). After adding the background variables as the control, the participants in the experimental group reported significant improvement in all mediator and outcome variables (ie, increased future time perspectives ($\beta=0.18$, $p<0.05$), reduced temporal discounting shortly ($\beta=-0.31$, $p<0.05$), reduced temporal discounting in the distance future ($\beta=-0.29$, $p<0.05$), improved financial attitudes ($\beta=0.21$, $p<0.05$) and better financial planning ($\beta=0.19$, $p<0.05$)). Notably, the standardized effect of our Python-based financial projection on temporal discounting was much larger than that of our previous personalized financial projection (−0.31 vs. −0.11), in which we did not present the projection design and coding to participants (ie, only showing them the projection outcomes in the mobile application).

Due to the limited sample size, we could not add background variables as controls in the structural model. Considering that the experiment's effects on two temporal discounting measures were significant only after adding control variables, we excluded them from the structural model as mediators. In the structural model, we tested the mediational effects of future time perspectives and financial attitudes on financial planning among participants. Based on the results shown in Figure 3, financial attitudes, rather than future time perspectives, were a significant mediator. Specifically, positive financial attitudes *fully* mediated the impact of Python-based financial projection on financial planning. After practicing with the Python-based financial projection, participants showed more positive financial attitudes ($\beta=0.24$, $p<0.05$), which in turn reminded them that they could further optimize their financial planning and increase their intention to plan for personal finance ($\beta=0.57$, $p<0.001$). After modelling the mediational effect, we noticed that the direct effect of the Python-based financial projection on financial planning disappeared.

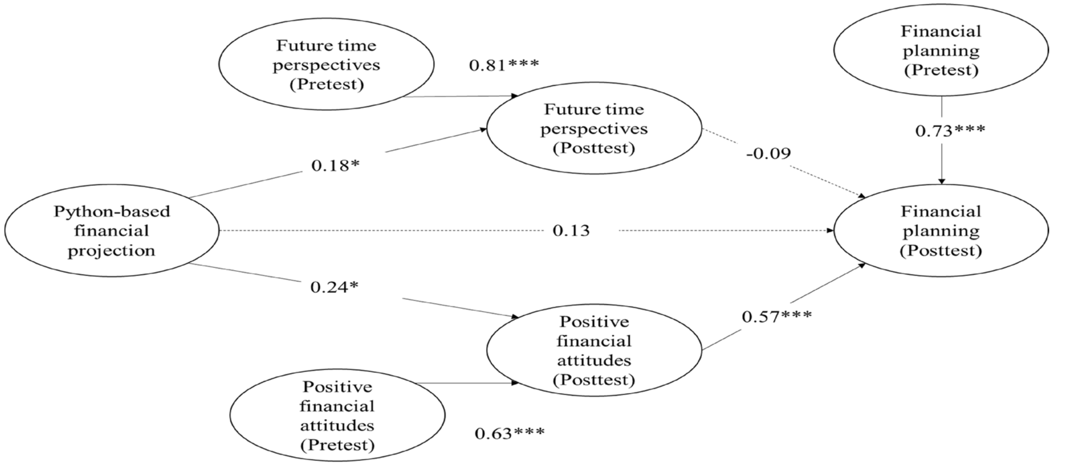

FIGURE 3 Results of experiment-based structural model. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

DISCUSSION

Research calls for upgrading standardized financial education, as its *generic* curriculum can only promote objective financial knowledge but is insufficient to change participants' underlying psychology and financial planning (Amagir et al., 2018; Kaiser & Menkhoff, 2020). Computer-based *personalized* financial projection could provide a solution, but our previous financial projection that masked its working mechanism had limited effects on changing the participants' underlying psychology (Zhu et al., 2023). In response, this study upgraded our previous financial projection by using Python to disclose its coding. We added Python-based personalized financial projections to the standardized financial education. Then, we conducted a randomized control trial among a sample of young working adults in Hong Kong to determine whether this newly added component could promote participants' financial planning by affecting their psychology (ie, promoting their future time perspectives and financial attitudes and reducing their temporal discounting).

We found that the Python-based personalized financial projection transformed the underlying psychology of participants expectedly and overwhelmingly in several ways. First, although both Python-based and our previous financial projection (masking the coding details) were found to decrease temporal discounting, the former's standardized effect was almost three times as large as that of the latter (ie, Hypothesis 1 is supported). Python-based coding presents the design of financial projections and elaborates on why current financial choices determine future financial status. The coding serves as a bridge to establish the participants' psychological connectedness to their future selves. When participants are well guided from the present to the future, the future becomes less obscure, more vivid and more valuable. In contrast, our previous financial projection masked the design of the projection program and only made the participants learn about their financial status covariates with their current choice. The specific mechanism that links the present to the future remains a black box.

Second, the Python-based financial projection was found to significantly promote future time perspectives (ie, Hypothesis 2 is supported), but the same effect was not found in our previous financial projection masking the coding (Zhu et al., 2023). Our previous projection only allows users to adjust personal inputs (eg, salary, expenditure, saving rate), while Python-based projection enables participants to adjust personal inputs, background economic parameters (eg, interest rate, salary growth, inflation), and even the logic of the projection program. In other words, the Python-based financial projection encourages participants to project their financial status in the future based on their perceptions of economic prospects in society. For the participant, the "the projected future" no longer means the future self alone but the future self in the future society. Python-based projection presents a more comprehensive simulation than the previous projection (that sets background parameters as default and does not allow participants to integrate their perceptions of the economic environment into the financial projection). Unsurprisingly, therefore, Python-based projection significantly promoted future time perspectives by taking participants to see a more vivid and realistic future. In contrast, our previous financial projection (masking the coding) failed to do so (Zhu et al., 2023).

Third, we found that Python-based financial projections could significantly promote the financial attitudes of participants, supporting Hypothesis 4. In our previous experiment, standardized financial education could not significantly shape financial attitudes (Zhu, 2020). Our previous investigation did not test the effect of personalized financial projection (masking the coding) on financial attitudes (Zhu et al., 2023). In this study, the Python-based financial projection maximized participants' opportunity to engage in "active experimentation" highlighted in the ELT (Kolb et al., 2014). Participants have the opportunity to engage in a multitude of experiments through the dynamic adjustment of individual inputs—such as the proportion of salary dedicated to savings and personal investment—as well as broader economic parameters like interest rates and inflation. The flexibility extends even to the code employed for projection. Within this framework, participants gain a comprehensive grasp of critical financial concepts, most notably compound interest. This understanding transcends different macroeconomic landscapes and diverse projection models tailored to individual life trajectories. In essence, this underscores the robustness of healthy financial practices—like consistent savings and prudent asset allocation for investment. These practices consistently contribute to the enhancement of future financial well-being, irrespective of an economy's macroeconomic fluctuations and the personalized course of one's life journey. In contrast, our preceding financial projection (masking the coding) limited participants to the realm of trial and error by solely adjusting personal inputs. Regrettably, this restricted approach may inadvertently propagate a misperception. Namely, participants might wrongly assume that prudent financial planning solely impacts long-term financial well-being within the confined scenario established by the projection system.

Finally, we stress that our Python-based financial projection significantly promoted financial planning among participants and that positive financial attitudes fully mediated this effect, indicating Hypothesis 5 is supported. The entire mediation well echoes the theoretical foundation of personalized financial projection that it affects the financial behaviours of participants by changing their underlying psychology (Bartels & Urminsky, 2015; Wiener & Doescher, 2008). Our findings justified adding Python-based personalized financial projections to standardized financial education. Future research may examine financial education composed of two components and determine whether it can simultaneously promote objective financial knowledge, financial attitudes and financial planning, and whether standardized financial education and Python-based financial projection interact to strengthen all effects.

While the Python-based financial projection training does encompass a substantial time investment—a comprehensive 5-hour instruction on Python basics followed by an additional 2-hour training session focused on manipulating projection code—we firmly assert that this allocation of time and resources proves exceptionally efficient. This efficiency stems from the profound and wide-ranging positive impacts that extend beyond the confines of the outcome variables we directly measured. The personalized projection serves as a vivid testament to the immense power of Python within the realm of financial management. This exposure often acts as a catalyst, motivating participants to delve deeper into the realm of

Python's capabilities. Some may be inspired to explore its applications in diverse financial calculations and planning scenarios, thereby broadening their horizons. It is not inconceivable that participants could experience a transformation in their approach to understanding the financial landscape once they harness Python's prowess to discern the consequences of various financial activities.

In an era marked by pervasive digitization, where programming and algorithms drive enhanced production efficiency, familiarity with Python basics emerges as a definite advantage. The newfound grasp of Python programming might even spur some participants to contemplate its applicability beyond the financial realm. This could potentially extend to assisting with the execution of their occupational responsibilities, tapping into the versatility of coding to streamline their professional tasks. In essence, despite the investment of time, the Python-based financial projection training acts as a gateway to a realm of practical skills and insights that have the potential to resonate far beyond the immediate training period.

LIMITATIONS

Two limitations should be stressed and both motivate future research to continue investigating the impact of Python-based financial projections on multidimensional financial literacy. First, our data did not show the significant mediational role of future time perspectives (ie, Hypothesis 3b is not supported), which might be due to the imperfect design of the data collection. With the two-wave data, the “effects” of mediators on the outcome variable are the cross-sectional associations. Adding a third wave may alter the findings and report the significant effect of the future time perspective on financial planning. Second, we did not test the mediational role of temporal discounting in financial planning (ie, Hypothesis 3a was not examined) due to the limited sample size. Our findings showed that Python-based financial projections significantly reduced temporal discounting after controlling for background variables. Unfortunately, however, we could not add background variables to the structural model due to an inadequate degree of freedom with a small sample size. Extending the sample size in future research could be a solution.

CONCLUSIONS

This study is the first worldwide to integrate Python into personal finance education. We conducted a randomized control trial to examine the effectiveness of Python-based financial projection (experiment group: standardized financial education plus Python-based financial projection; control group: standardized financial education). We found that Python-based financial projection promoted future time perspectives, financial attitudes and financial planning of young working adults in Hong Kong and reduced their temporal discounting. Positive financial attitudes fully mediated the positive effect of Python-based financial projections on financial planning. We also concluded that exposing the coding to participants strengthened the effect of personalized financial projection by comparing our results to those in our previous investigation. Inspired by these findings, we call for more Python-based themed financial projections (eg, housing planning and health planning) and believe they may benefit the overall financial planning of working adults. More importantly, our findings suggest upgrading standardized financial education (ie, combining standardized financial education with Python-based personalized financial projection). The former promotes objective financial knowledge, while the latter works by changing underlying psychology and affecting financial planning.

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CONFLICT OF INTEREST STATEMENT

The author declares no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The ethical approval was obtained from the Human Research Ethics Committee (HREC) of The Lingnan University (Hong Kong) before the data collection.

INFORMED CONSENT

Informed consent was obtained from all individual participants included in the study.

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